

ELVIS NIYOMWUNGERE

Software Architect & Full-Stack Developer | AI & Computer Vision

CONTACT

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CORE SKILLS

Architecture & Backend
Software Architecture, Enterprise Systems, Multi-tenant Architecture, REST APIs, PHP/Laravel, Python/FastAPI, WebSockets, Distributed Systems

Frontend
React/Next.js, Vue.js, JavaScript/TypeScript, Tailwind CSS

Databases
PostgreSQL, MySQL, NoSQL

AI & Computer Vision
Artificial Intelligence, Computer Vision, Object Detection, OCR, ANPR/LPR, Video Analysis

Security & Infrastructure
Cybersecurity, RBAC & Access Control, Linux/Ubuntu Server, Nginx, Deployment, System Monitoring, Android MDM

LANGUAGES

French (Advanced), English (Advanced), Kiswahili (Intermediate), Kirundi (Fluent)

PROFESSIONAL SUMMARY

Software architect and full-stack developer with 3+ years of experience designing secure, scalable digital systems, with expertise in artificial intelligence, Computer Vision, and multi-tenant architectures. Experienced in SaaS platform design, API integration, database management, and cybersecurity, including enterprise and government projects. Passionate about technological innovation tailored to institutional and economic needs. Open to remote opportunities.

WORK EXPERIENCE

Office of the Prime Minister Feb 2026–Present
Architect & Digital Systems Developer

- Design and develop digital systems for the digitalization of administrative and financial processes, based on analysis of functional and technical requirements and software architectures tailored to institutional needs.
- Develop and integrate business features, validation workflows, and user, role, and permission management.
- Design and manage databases and data processing and securing mechanisms.
- Implement solutions improving traceability, centralization, and efficiency of administrative processes.
- Deploy, maintain, and continuously improve applications and their technical infrastructure.
- Contribute to the evolution of the architecture and to resolving technical issues related to institutional digital systems.

BeYou Digital Solutions Oct 2025–Jan 2026
Founder & Lead Developer

- Designed and developed enterprise software architectures for multi-user, multi-tenant, and multi-sector systems, turning business requirements into structured, secure, and scalable digital solutions.
- Built web platforms and management systems integrating REST APIs, databases, business workflows, authentication, RBAC, and traceability.
- Backend development with PHP/Laravel and Python/FastAPI; modern interfaces with React/Next.js, Vue.js, JavaScript/TypeScript, and Tailwind CSS.
- Designed and integrated AI and Computer Vision solutions (object detection, OCR, ANPR/LPR, video analysis, inference pipelines) and structured AI platforms covering model management, versioning, deployment, monitoring, and performance analysis.
- Built real-time communications via WebSockets, video streams, and specialized backend services.
- Implemented application security: authentication, access control, organization isolation, and data protection.
- Designed and managed PostgreSQL and MySQL databases (modeling, migrations, optimization, multi-tenant data); deployed and administered Linux/Ubuntu Server environments and Nginx, with backup and maintenance mechanisms; configured Android devices and centralized device management (MDM).
- Audited, secured, and improved existing management systems with a focus on performance, reliability, and scalability; led projects technically, coordinated development, and monitored emerging technologies in Software Engineering, AI, and Computer Vision.

Freelance Apr 2024–Sep 2025
Full Stack Developer

- Full development of Web and Mobile applications
- Design of secure architectures
- Management of relational and NoSQL databases
- Performance optimization and strengthening of application security

Freelance May 2023–Mar 2024

Web & Mobile Developer

- Design and development of professional websites (showcase site, online store, portfolios, etc.)
- Development of dynamic web applications
- API integration and database management
- Direct collaboration with clients (needs analysis, maintenance)

ARCT (Agence de Régulation et de Contrôle des Télécommunications) Apr 2021–Jul 2021

Academic Intern

- Analyzed institutional information systems
- Provided technical assistance and contributed to internal IT activities
- Delivered initiation training on ICT infrastructures
- Analyzed telecommunications regulatory data to support compliance monitoring and reporting
- Collaborated with IT teams to document network architecture and identify potential security enhancements
- Conducted research on emerging ICT technologies to inform strategic planning initiatives
- Assisted in the development of user guides for internal software applications to improve staff training

PROJECTS

GovTech eCivilBurundi & eID Project — GovTech R&D

- Developed fraud detection and prevention mechanisms
- Designed interoperability mechanisms between institutions

BeYou Nexus — Multi-Sector SaaS Platform

- Modules for Education, Banking, Enterprise, Hospitals, and Agriculture
- Centralized and secured data, advanced management of users, roles, and permissions, and multi-tenant isolation
- Modular architecture enabling progressive evolution of the platform

SmartVisionLab — Artificial Intelligence & Computer Vision Platform

- Multi-tenant, multi-organizational AI platform covering model lifecycle management, deployment, supervision, inference, and monitoring
- Computer Vision and video analysis: object detection, OCR, ANPR/LPR, and license plate analysis
- Camera and video stream management, with real-time events and operations via WebSocket
- Operational intelligence dashboards
- Security, RBAC, audit, and governance of AI platforms

PORTAIX — Smart Port & Intelligent Port Management Concept

- Intelligent management of vehicles and cargo, with barrier control and automation
- Automatic vehicle identification (ANPR/LPR), OCR, and Computer Vision integration in port operations
- Intelligent weighing and weighbridge integration
- Billing automation and integration with payment systems
- Interoperability with port and customs systems, operational supervision, and data analysis

EDUCATION

University of Burundi, Bujumbura, East Africa, Burundi 2026
Bachelor of Science in Computer Science in Information Technology

Ets Kamenge, Bujumbura, East Africa, Burundi 2021
Diploma A2